Interface Orbital Magnetism in Ti/Ni and Cr/Ni Bilayers Relevant to Spin–Orbit-Torque Materials

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Spin–orbit-torque materials require coordinated control of angular-momentum generation, interfacial transmission, and magnetic-layer response. Recent orbitronics studies have identified orbital angular momentum as an additional current-induced degree of freedom in light transition-metal heterostructures. Here we investigate epitaxial Ti/Ni and Cr/Ni bilayers as 3d-metal/Ni interface systems relevant to SOT materials. Ti and Cr are selected as representative 3d source layers with reported orbital Hall responses, while Ni provides a thin ferromagnetic layer whose spin and orbital moments can be resolved element specifically. Temperature- and angle-dependent Ni $L_{2,3}$-edge x-ray magnetic circular dichroism reveals clear dichroic contrast and systematic angular evolution in Ti/Ni, enabling analysis of Ni orbital magnetism and orbital-moment anisotropy. This element-specific description provides a microscopic reference for interpreting thickness- and symmetry-dependent torque signatures from ST-FMR and harmonic Hall measurements.

***Index Terms*—orbital Hall effect, interface orbital magnetism, spin–orbit torque, XMCD, Ti/Ni, Cr/Ni, ST-FMR, harmonic Hall.**

I. Orbital Interfaces in SOT Material Stacks

Spin–orbit-torque magnetic memory has motivated extensive searches for material stacks that combine efficient angular-momentum transfer, low damping, and stable magnetic interfaces [1]. Heavy metals such as Pt, Ta, and W remain central to SOT materials, but their performance is determined by spin Hall efficiency, interfacial transparency, magnetic-layer damping, and materials compatibility. Orbitronics extends this framework by introducing orbital angular momentum as a current-induced channel for angular-momentum transport and conversion [2]–[6].

Ti and Cr have both become important 3d transition metals in orbital Hall studies. Current-induced orbital accumulation has been directly observed in Ti, while Cr has been reported to exhibit a sizable orbital Hall response [3], [4]. In this study, Ti and Cr are used to form epitaxial *X*/Ni interfaces with different 3d electronic environments. The focus is on how the adjacent 3d metal modifies the spin and orbital magnetic state of the Ni layer.

Recent studies show that orbital-torque measurements in metallic bilayers are influenced by source-layer orbital response, interface transparency, orbital-to-spin conversion, and magnetic-layer properties [5]–[10]. Element-specific XMCD directly probes the spin and orbital moments of Ni, providing a microscopic basis for connecting interface orbital magnetism with electrical torque readout.

II. Epitaxial 3d-Metal/Ni Interface Platform

The films are grown by molecular beam epitaxy on annealed MgO(001) substrates. Reflection high-energy electron diffraction is used before metal deposition to assess the crystalline quality of the substrate surface. The heterostructures have the nominal form

$$\mathrm{MgO(001)}/X(t_X)/\mathrm{Ni}(t_{\mathrm{Ni}})/\mathrm{cap},$$

where X = Ti or Cr, and t_X denotes the Ti or Cr source-layer thickness. The Ti and Cr layers are varied within the nanometer-thickness regime, while Ni is maintained as a thin ferromagnetic layer sensitive to the adjacent interface. This geometry enables the orbital magnetism of Ni to be examined as a function of the neighboring 3d metal and *X*/Ni interface.

Epitaxial growth is important because interfacial symmetry, roughness, orbital hybridization, and electronic transparency can affect angular-momentum transfer across metallic interfaces [7], [8]. The Ti/Ni and Cr/Ni bilayers therefore provide two epitaxial 3d-metal/Ni interface systems for evaluating interface-modified orbital magnetism in Ni.

III. Temperature- and Angle-Dependent XMCD of Ni

Ni $L_{2,3}$-edge XMCD is used to extract the spin moment m_S, orbital moment m_L, and orbital-to-spin moment ratio m_L/m_S using XMCD sum-rule analysis [11], [12]. Measurements at 80 K, 150 K, and 300 K probe the thermal evolution of the Ni magnetic state. Angle-dependent XMCD, acquired from near-normal to tilted geometries, probes the anisotropy of the Ni orbital moment and its dependence on the adjacent 3d layer.

Representative MgO/Ti/Ni/cap spectra measured at 80 K show well-resolved Ni L_3 and L_2 absorption features and a clear XMCD signal, $\mu^{-}-\mu^{+}$, over multiple incidence angles, as shown in Fig. 1. The dichroic contrast exhibits a systematic angular evolution as the field/x-ray geometry is varied with respect to the film normal, indicating a pronounced angle-dependent Ni magnetic response. This angular evolution provides the experimental basis for evaluating orbital-moment anisotropy and for comparing m_L, m_S, and m_L/m_S across Ti/Ni and Cr/Ni interfaces.

After normalization to the isotropic XAS background, the integrated dichroic intensities at the L_3 and L_2 edges provide quantitative access to the Ni spin and orbital moments. The angular dependence of m_L/m_S, together with the evolution of the XMCD line shape and magnetic contrast, serves as a descriptor of interface-modified orbital magnetism. Such information is particularly relevant for ultrathin ferromagnets, where orbital-moment anisotropy is closely connected to spin–orbit coupling and magnetic anisotropy [13]. Magneto-optical Kerr measurements provide auxiliary information on hysteresis and domain behavior, while the quantitative moment analysis is based on element-specific XMCD.

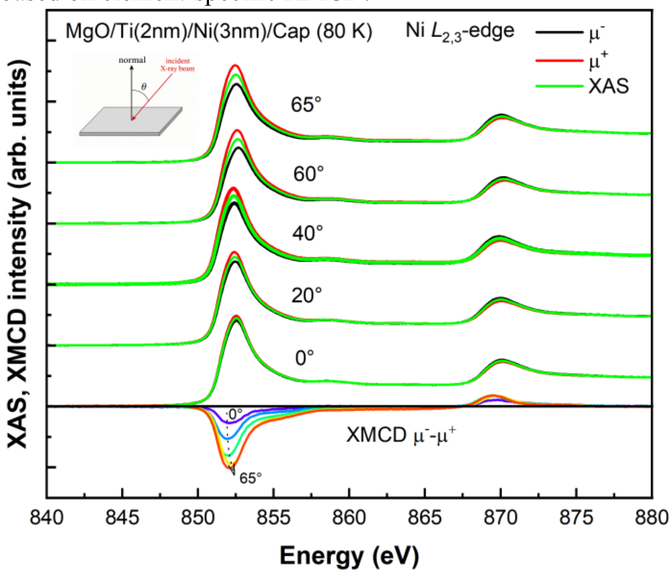

Fig. 1. Angle-dependent Ni $L_{2,3}$-edge XAS and XMCD spectra of MgO/Ti(2 nm)/Ni(3 nm)/cap measured at 80 K. The incidence angle θ is defined with respect to the film normal. The upper spectra show μ^{-}, μ^{+}, and averaged XAS for different angles, while the lower spectra show the corresponding XMCD difference, μ^{-}-μ^{+}. The clear dichroic contrast at the Ni L_3 and L_2 edges and its systematic angular evolution provide element-specific sensitivity to the Ni magnetic response and orbital-moment anisotropy.

IV. Torque Signatures and Interface-Orbital Analysis

The electrical response is analyzed using spin-torque ferromagnetic resonance and second-harmonic Hall measurements. These measurements provide damping-like and field-like torque components, while angular dependence and source-layer variation help identify source-layer and interface-sensitive trends. Because changing $t_X$modifies the conductance of the Ti or Cr layer, current redistribution across X, Ni, and the cap layer is included in the torque-efficiency analysis.

To guide the interpretation of electrical measurements, the torque response can be organized phenomenologically as

$$\xi(t_X) = \xi_{source}(t_X) + \xi_{int} + \xi_{Ni},$$

where $\xi_{source}(t_X)$ denotes spin or orbital Hall contributions associated with the Ti or Cr layer, ξ_{int} represents interface-related orbital accumulation, orbital Rashba–Edelstein response, or interfacial conversion, and ξ_{Ni} accounts for magnetic-layer contributions. This representation supports comparison of torque symmetry, sign, and source-layer dependence in relation to the XMCD-derived Ni orbital state.

This combined approach examines how a thin Ni ferromagnetic layer responds magnetically and dynamically when interfaced with different 3d orbital-source layers. A source-layer contribution is expected to exhibit a systematic dependence on t_X, whereas interfacial and magnetic-layer terms may appear through finite-thickness offsets, field-like components, or changes correlated with m_L/m_S and orbital-moment anisotropy. Correlating torque signatures with Ni orbital magnetism therefore provides a route to evaluate orbital-interface effects in epitaxial 3d-metal/Ni heterostructures. This approach links element-specific interface orbital magnetism to torque readout in 3d-metal/Ni bilayers and provides a materials-level basis for assessing light-metal angular-momentum sources in SOT stacks.

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